

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	CSM-A13
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Skill Recorder

Abstract

Repetitive computer-based tasks are common in academic, professional, and administrative work. Users often perform the same sequence of actions across web browsers, desktop applications, and other digital environments. Although automation can reduce this repetitive effort, creating an automation workflow usually requires technical knowledge of scripting, automation tools, or application-specific configurations. This makes automation difficult for users who can perform a task manually but do not have the expertise to describe and implement the corresponding automation. This project proposes an AI-assisted Skill Recorder that converts demonstrated user workflows into structured and reusable skills. The system records the user's work session and captures relevant information about the actions performed during the task. The recorded session is then analyzed to understand the user's intent and identify the sequence of steps involved. Based on this analysis, the system generates a structured representation of the workflow that can be transformed into a reusable skill or automation. This approach allows users to demonstrate how a task is performed instead of manually defining every step required for automation. Existing screen-recording and macro-based automation approaches primarily focus on replaying predefined user actions. Such approaches can become difficult to maintain when the workflow or application interface changes. The proposed system focuses on understanding the workflow at a higher level by combining user activity recording with AI-based analysis. This can make the creation of reusable automation more accessible and reduce the effort involved in defining repetitive workflows. The project primarily focuses on recording and analyzing supported desktop workflows and generating reusable skill representations. It does not aim to automate every type of application or guarantee autonomous execution of arbitrary tasks. The effectiveness of the system will be evaluated by measuring its ability to accurately reconstruct the user's intended task, preserve the correct order of workflow steps, and generate consistent reusable skills across representative workflows.

References

1.Xie, T., Zhang, D., Chen, J., Li, X., Zhao, S., Cao, R., Hua, T. J., Cheng, Z., Shin, D., Lei, F., Liu, Y., Xu, Y., Zhou, S., Savarese, S., Xiong, C., Zhong, V., & Yu, T. (2024). OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. Advances in Neural Information Processing Systems (NeurIPS).
DOI: 10.52202/079017-1650.

2.Cheng, K., Sun, Q., Chu, Y., Xu, F., Li, Y., Zhang, J., & Wu, Z. (2024). SeeClick: Harnessing GUI Grounding for Advanced Visual GUI Agents. Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL).
DOI: 10.18653/v1/2024.acl-long.505.

3.Zhang, C., Li, L., He, S., Zhang, X., Qiao, B., Qin, S., Ma, M., Kang, Y., Lin, Q., Rajmohan, S., Zhang, D., & Zhang, Q. (2025). UFO: A UI-Focused Agent for Windows OS Interaction. Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL).
DOI: 10.18653/v1/2025.naacl-long.26.

4.Rawles, C., Li, A., Rodriguez, D., Riva, O., & Lillicrap, T. (2023). Android in the Wild: A Large-Scale Dataset for Android Device Control. Advances in Neural Information Processing Systems (NeurIPS).
DOI: 10.52202/075280-2609.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A4228	Cheeraboyina Lokeswari	
2	23BQ1A4223	Bodimalla Vamsi Krishna Reddy	
3	23BQ1A4204	Akkinapalli Vignesh	
4	23BQ1A4257	Guddanti Venkata Sai Yateswar Ram	

Project Guide

Name: Dr. M. Pardha Saradhi

Signature: